SIMPLE CONTACT MANAGEMENT SYSTEM (MVP)

Product Requirements Document

Prepared By:
RISHIKESH S

Role:
Aspiring Product Manager

Technology Stack:
Java | MySQL | JDBC

Institution:
Rathinam College Of Arts and Science

Date:
24 - 02 - 2026

PRODUCT REQUIREMENTS DOCUMENT

1. Product Overview

The **Simple Contact Management System** is a console-based digital application developed using Java and MySQL with JDBC connectivity.

This product is designed to help users efficiently store, manage, and retrieve contact information in a structured digital format.

The project was built as a **Minimum Viable Product (MVP)** to solve real-world problems related to manual contact storage.

Additionally, this product represents a practical product-building journey — from identifying a real user problem to designing and launching a functional solution.

2. Problem Statement

Many individuals still store contact details manually using notebooks, files, or unorganized digital notes.

This leads to several issues:

- Important contacts may get lost or forgotten
- Manual records can be damaged easily
- Information is often stored in multiple places
- Updating existing data becomes difficult
- Accessing contacts during urgent situations is inconvenient

These challenges highlight the need for a simple digital system to manage contact information securely and efficiently.

3. Why This Product Was Chosen

This project was selected based on academic, practical, and product-thinking motivations.

From an academic perspective:

- The curriculum required building a real-world management system using Java and MySQL connectivity.
- The project provided hands-on experience with JDBC database integration.

From a real-world perspective:

- Contact management is a common and essential everyday need.
- Digital storage offers better reliability compared to manual record-keeping.

From a product mindset perspective:

- This project served as an opportunity to build a first real product.
- It enabled experience in identifying a real problem and creating a user-focused solution

---

4. Target Users

This system is designed for:

- Individuals who want to digitally store contacts
- Users who frequently manage contact information
- People needing quick access to emergency contacts
- Anyone seeking a simple alternative to manual storage

---

5. Product Vision

To develop a simple digital platform that eliminates problems associated with manual contact storage by enabling users to securely store, manage, and access contact details anytime.

---

6. MVP Features

The Minimum Viable Product includes the following core features:

Add Contact

Allows users to store new contact details such as name, phone number, email, and address.

View Contacts

Enables users to retrieve and view stored contact information.

Update Contact

Allows users to modify existing contact details.

Delete Contact

Enables removal of unwanted contacts from the system.

---

7. Future Enhancements

Planned improvements for the web-based version include:

- User login and authentication system
- Cloud backup functionality
- Multi-device accessibility
- Contact sharing options
- Advanced search and filtering features

---

8. Product Development Lifecycle

Conceive

Identified the real-world problem of manual contact storage.

Plan

Selected the contact management solution and defined core features.

Build

Developed a console-based application using Java and MySQL.

Test

Validated database connectivity and tested all CRUD operations.

Launch

Published the project on GitHub for public access and demonstration.

Future Growth

Planning to develop a full web-based version based on feedback.

---

9. Product Lifecycle Stages

Birth Stage

Idea validation and MVP development.

Growth Stage

User adoption increases after GitHub release and future web deployment.

Maturity Stage

Product stabilizes with enhanced features and consistent usage.

Decline Stage

May occur if the product is not updated or replaced by advanced solutions.

---

10. Product Roadmap

1. Setup development environment and tools
2. Establish Java-MySQL database connectivity
3. Develop individual CRUD features
4. Integrate features into a menu-driven system
5. Test complete functionality
6. Publish project on GitHub
7. Create PRD documentation
8. Collect user feedback
9. Develop full web-based application

---

11. Success Metrics (KPIs)

Product success will be measured using:

- Number of contacts stored in the system
- Frequency of CRUD feature usage
- User feedback and satisfaction

These indicators help evaluate whether the product delivers meaningful value to users.

---

12. Conclusion

The Simple Contact Management System demonstrates the complete process of solving a real-world problem through structured product development.

It serves as a strong foundation for future expansion into a scalable web-based application.

---

13. Application Screenshots

This section displays the key interfaces of the Simple Contact Management System MVP application.

Main Menu Interface

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                                    java

PS C:\Users\Admin\simple contact management> javac -cp "mysql_jarconnector/mysql-connector-j-8.1.0.jar;src_folder" src_folder/Contact_Management_System.java
PS C:\Users\Admin\simple contact management> java -cp "mysql_jarconnector/mysql-connector-j-8.1.0.jar;src_folder" Contact_Management_System

=====  Contact Management Menu =====
1. Add Contact
2. View Contacts
3. Update Contact
4. Delete Contact
5. Exit
Enter your choice: 1
Connected to Database
Enter Name: raju
Enter Phone: 9894567418
Enter Email: raju1234@gmail.com
Enter Address: tirupur,tamilnadu
 Contact Added Successfully!
```

Add Contact Feature

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                                    java

PS C:\Users\Admin\simple contact management> javac -cp "mysql_jarconnector/mysql-connector-j-8.1.0.jar;src_folder" src_folder/Contact_Management_System.java
PS C:\Users\Admin\simple contact management> java -cp "mysql_jarconnector/mysql-connector-j-8.1.0.jar;src_folder" Contact_Management_System

=====  Contact Management Menu =====
1. Add Contact
2. View Contacts
3. Update Contact
4. Delete Contact
5. Exit
Enter your choice: 1
Connected to Database
Enter Name: raju
Enter Phone: 9894567418
Enter Email: raju1234@gmail.com
Enter Address: tirupur,tamilnadu
 Contact Added Successfully!
```

View Contacts Feature

```
=====  Contact Management Menu =====
1. Add Contact
2. View Contacts
3. Update Contact
4. Delete Contact
5. Exit
Enter your choice: 2

---- Contact List ----
ID: 2
Name: raj
Phone: 75432578990
Email: raj123@gmail.com
Address: covai,tamilnadu
-----------------------
ID: 3
Name: raju
Phone: 9894567418
Email: raju1234@gmail.com
Address: tirupur,tamilnadu
-----------------------
```

Update Contact Feature

```
=====  Contact Management Menu =====
1. Add Contact
2. View Contacts
3. Update Contact
4. Delete Contact
5. Exit
Enter your choice: 3
Enter Contact ID to Update: 3
Enter New Name: raju
Enter New Phone: 9894567418
Enter New Email: raju1234@gmail.com
Enter New Address: erode,tamilnadu
 Contact Updated Successfully!
```

Delete Contact Feature

```
=====  Contact Management Menu =====
1. Add Contact
2. View Contacts
3. Update Contact
4. Delete Contact
5. Exit
Enter your choice: 4
Enter Contact ID to Delete: 3
 Contact Deleted Successfully!

=====  Contact Management Menu =====
1. Add Contact
2. View Contacts
3. Update Contact
4. Delete Contact
5. Exit
Enter your choice: 2

---- Contact List ----
ID: 2
Name: raj
Phone: 75432578990
Email: raj123@gmail.com
Address: covai,tamilnadu
-----------------------
```